

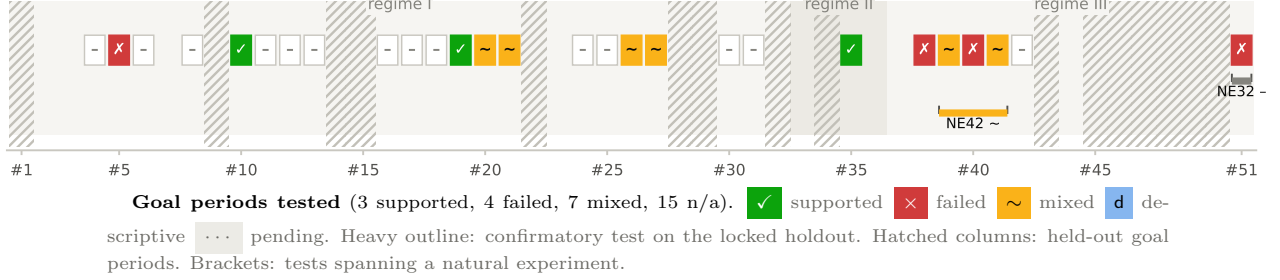
H48: The settling time is a mixing time on the read-out graph

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Agents read each other within minutes but content settles over hours, so settling is not "everyone has read everyone"; whether it scales with a slower read-out time could not be tested at the available noise.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH173 claimed content settles after a kickoff once every agent has read every other agent's work: a bulk mixing time on the read-out graph, not whole-graph λ_2 . Using DQ1's context ledger we measured, per goal period, the time until 90% of room-mate pairs had read each other (T_{90}) and the mixing time of the read-out chain. We compared these with the kickoff-remanence settling time τ (1-h bins, two embedding models) on 27 non-holdout periods, plus four native tests (NE42, #38, #51 joins, NE32). Coverage takes minutes (T_{90} median 0.24 active h); settling takes hours (τ median 4.5 h, about $10\times$). Neither predictor beats a constant across periods (LOPO gains -0.01 to -0.14). But τ is detectable in only 11/27 periods, and a noise-calibrated synthetic has no power. **Inconclusive**; the shallow-coverage version fails on magnitude. Holdout script written, not run.

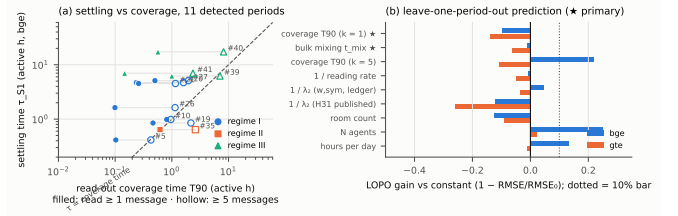
PHYSICAL MODEL

Vector spins (model 11) updated only at read-outs. Agent i 's content $x_i \in \mathbb{R}^{32}$ (whitened statement mean) is quenched toward the kickoff target $\hat{k}$ (H54). At each call it averages toward the messages that newly entered its context (ledger batch B), with weight α . W_{ij} : j 's messages read by i per active hour; u_i : reading-turn rate; Q : the read-out chain. Rivals: a field clock (kickoff decays on its own), the agent's own-call clock, the day clock, and λ_2 (H31).

$$x_i \leftarrow (1 - \alpha) x_i + \alpha \langle x_{s(m)} \rangle_{m \in B}, \quad Q_{ij} = u_i \frac{W_{ij}}{\sum_k W_{ik}} - \delta_{ij} u_i$$

$$C_k(t) = \Pr_{(i,j)} [n_{ij}(t) \geq k], \quad t_{\text{mix}}^{\text{bulk}} = \text{med}_i \min \{t : \text{TV}(e_i e^{Q^t}, e_i e^{Q^\infty}) \leq \frac{1}{4}\}$$

$$A_{\text{ex}}(t) = \langle \cos(v_i, \hat{k}) \rangle_i - \langle \cos(v_i, \hat{k}_q) \rangle_{i,q} = A_\infty + (A_0 - A_\infty) e^{-t/\tau}$$



Observables. (a) Settling time against coverage time, 11 periods with detected settling. Filled: 90% of pairs read once; hollow: read five times. Settling sits about $10\times$ above single-read coverage (dashed line $\tau = T_{90}$) and near depth-5 coverage (post hoc). (b) Leave-one-period-out gain over a constant. Both primaries (stars) lose in both models; the bge-only leads (depth-5 coverage, N) do not replicate with gte.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

One useful negative fact for operators: in broadcast rooms, reading is not the bottleneck. Every agent has seen every room-mate within minutes of a kickoff, yet content keeps relaxing for hours. So merging rooms or raising call rates will not speed convergence on a new goal by giving agents more to read; if read-out drives settling at all, each read moves an agent by about $1/250$. Coverage times computed from the ledger are cheap, but they are weakest-link numbers: one silent agent sets a whole room's coverage. The scaling hypothesis itself is unanswered, because per-period settling times are too noisy and too varied in shape to rank 27 periods. As posed, this is a measurement problem more than a finding.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 T_{90} or t_{mix} beats constant, both models	gains $-0.10/-0.14, -0.01/-0.06$; 11 detected (< 12)	inconclusive
P2 $\tau/T_{90} \in [1/3, 3]$	10.1 (IQR 2.9–18.6); gte 4.5	failed
P3 bulk beats worst case	0/3 pairs	failed
P4 holds within regime	gains $-0.08, -0.12$	failed
P5, P6 robust to model, DQ5	ρ 0.80, 0.97	held
P8 settling detected $\geq 60\%$	41% bge, 52% gte	failed
P10 no λ_2 beats constant	best gain 0.05	held
NE42 τ ordered as T_{90}	ρ 0.30 bge, 0.90 gte (n 5)	mixed
#38 slow-covering room settles later	settles faster (gte τ_q 1.6 vs 12.3 d)	failed
#51 newcomers assimilate	3/11; they individuate	failed

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G10 $\checkmark$ supported
G19 $\checkmark$ supported
G20 $\sim$ mixed
G21 $\sim$ mixed
G26 $\sim$ mixed
G27 $\sim$ mixed
G35 $\checkmark$ supported
G38 $\times$ failed
G39 $\sim$ mixed

+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

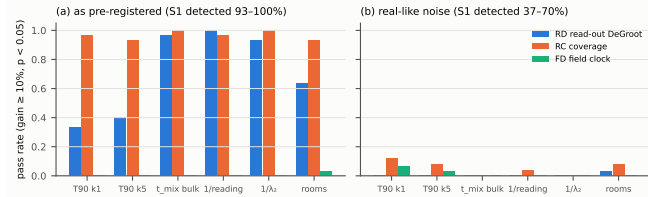
The settling estimate is fragile. A single exponential fits in about half the periods; the rest show fast drops then recovery, multi-day declines, or no kickoff signal (#51's private goals). H20's day-scale τ_q exists only in 10 long periods. The bge and gte detected sets overlap in 9 of 16 periods. Coverage counts chat messages only, not artifacts. Regime-I read times use low-confidence call starts. 19 predictors $\times$ 2 models were looked at; only P1 is multiplicity-controlled, so the bge leads are leads. The real-like calibration was chosen after seeing the data.

ROUND 2

Round 2: estimate the per-read pull α at message level (H29-style, on ledger read-outs) and predict $\tau = 1/(\alpha u)$ directly; build a hierarchical settling estimator that recovers planted τ at real noise; test depth-5 coverage on the holdout.

- **H48-R1** Message-level α .
- **H48-R2** A settling estimator with power.
- **H48-R3** Depth-5 coverage (post hoc lead).
- **H48-R4** Newcomer individuation (from G51).

H48-settling-mixing-time · generated 2026-10-04 11:10 by infra/summaries/build_summaries.py · card hypotheses/H48-settling-mixing-time/README.md · ratings: H48 agent (suggested) (page 2)



Second figure. Synthetic: vector spins on the real read-out schedules. (a) At the pre-registered noise level, bulk predictors identify read-out truths (RD, RC) with no false positives under a field clock (FD). (b) At noise matched to the real detection rate (post hoc), no predictor passes under any truth, so the real-data null says nothing about scaling.